Derivation of the Hybrid Forcing Term Computation

Not Reviewed Yet

Let E_j denote the total contribution of the non-homogeneous forcing term for a specific frequency j at time t_k:

$$E_j = e^{-j^2 t_k} \int_0^{t_k} e^{j^2 s} \hat{F}_j(s)\, ds \tag{1}$$

To optimize the computation while maintaining rigorous accuracy, we partition the infinite series $\sum_{j=1}^{n_k} E_j$ into two phases using a threshold index N_{exact}.

Phase 1: The Exact Part ($j \leq N_{\text{exact}}$)

For low frequencies, the exponential term $e^{j^2 s}$ does not exhibit extreme growth. The integral's behavior across the entire interval $[0, t_k]$ is significant and is computed using standard numerical quadrature:

$$E_j^{(\text{exact})} = e^{-j^2 t_k} \int_0^{t_k} e^{j^2 s} \hat{F}_j(s)\, ds \tag{2}$$

Phase 2: The Asymptotic Part ($j > N_{\text{exact}}$)

For high frequencies, $e^{j^2 s}$ grows violently, localizing the dominant contribution of the integral entirely to the upper boundary $s = t_k$. We isolate this behavior using integration by parts. Let:

$$u = \hat{F}_j(s) \implies du = \hat{F}_j'(s)\, ds$$
$$dv = e^{j^2 s}\, ds \implies v = \frac{e^{j^2 s}}{j^2}$$

Applying the integration by parts formula yields:

$$\begin{aligned}\int_0^{t_k} e^{j^2 s} \hat{F}_j(s)\, ds &= \left[\frac{e^{j^2 s}}{j^2} \hat{F}_j(s)\right]_0^{t_k} - \int_0^{t_k} \frac{e^{j^2 s}}{j^2} \hat{F}_j'(s)\, ds \\ &= \left(\frac{e^{j^2 t_k}}{j^2} \hat{F}_j(t_k) - \frac{1}{j^2} \hat{F}_j(0)\right) - \int_0^{t_k} \frac{e^{j^2 s}}{j^2} \hat{F}_j'(s)\, ds\end{aligned}$$

Distributing the outside multiplier $e^{-j^2 t_k}$ across these three terms gives:

$$E_j = \underbrace{e^{-j^2 t_k}\left(\frac{e^{j^2 t_k}}{j^2}\hat{F}_j(t_k)\right)}_{\text{Term A}} - \underbrace{e^{-j^2 t_k}\left(\frac{1}{j^2}\hat{F}_j(0)\right)}_{\text{Term B}} - \underbrace{\frac{1}{j^2}\int_0^{t_k} e^{-j^2(t_k-s)}\hat{F}_j'(s)\,ds}_{\text{Term C}}$$
$$= \frac{\hat{F}_j(t_k)}{j^2} - \frac{e^{-j^2 t_k}}{j^2}\hat{F}_j(0) - \frac{1}{j^2}\int_0^{t_k} e^{-j^2(t_k-s)}\hat{F}_j'(s)\,ds$$

Taking the limit as $j \to \infty$, we analyze the decay rates:

1. **Term B** vanishes exponentially. Because $t_k > 0$, the exponent $-j^2 t_k$ drives the term to zero instantly.
2. **Term C** is mathematically negligible. A secondary application of integration by parts extracts an additional j^{-2} factor, meaning Term C shrinks at a rate of $\mathcal{O}(j^{-4})$.
3. **Term A** shrinks at a rate of $\mathcal{O}(j^{-2})$ and thus strictly dominates the sequence.

Therefore, the asymptotic behavior is entirely captured by Term A, collapsing the computationally expensive integral into a simple algebraic evaluation:

$$E_j^{(\text{asymptotic})} \approx \frac{\hat{F}_j(t_k)}{j^2} \tag{3}$$

Final Hybrid Algorithm

By combining the exact and asymptotic derivations, the total forcing summation can be computed efficiently while bypassing integration overflow for large n_k:

$$\text{Total Forcing} \approx \sum_{j=1}^{N_{\text{exact}}} \left(e^{-j^2 t_k}\int_0^{t_k} e^{j^2 s}\hat{F}_j(s)\,ds\right) + \sum_{j=N_{\text{exact}}+1}^{n_k} \frac{\hat{F}_j(t_k)}{j^2} \tag{4}$$